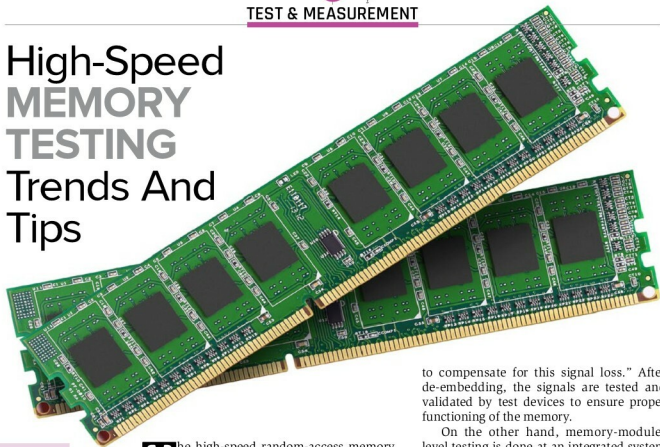


High-Speed MEMORY TESTING Trends And Tips



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The high-speed random-access memory (RAM) market is dominated by double-data-rate (DDR) technology with DDR4 ruling the pack. Mobile devices like tablets, laptops and smartphones use a low-power version of DDR (LPDDR) to lower their power consumption.

DDR memory is based on a parallel-line technology that uses a large number of channels to carry the data. Operating on such a large number of data lanes at a high speed presents its own challenges. This is driving upgradation of the respective test and measurement (T&M) devices.

Trends in memory testing

DDR chips are tested at wafer-probe and memory-module levels. Probing is an important aspect of catching the right signals of the memory by the test equipment. But the process of probing comes with many difficulties. To explain simply, when you use probes for DDR, there is a loss of signal. Such challenges call for a process called de-embedding.

Sanchit Bhatia, application expert in high-speed digital applications, Keysight Technologies, says, "Specialised ball-grid-array probes are the preferred choice. These are soldered between the memory and the board, DDR4 has an average speed of 3.2 gigabits per second. At such a high speed, there is signal loss when using these probes. De-embedding is done

to compensate for this signal loss." After de-embedding, the signals are tested and validated by test devices to ensure proper functioning of the memory.

On the other hand, memory-module-level testing is done at an integrated system level when all the component-level tests are completed.

Bhatia further explains that DDR testing can be done in two broad categories. The first type is parametric testing, which tests the integrity of DDR signals by analysing the timing of DDR waveforms and all other physical-layer parameters. The other category is functional or protocol testing, which tests whether the data is flowing as per the rules set. It evaluates whether all the commands are in place, addresses are correct and the signals are following all the protocols properly.

For DDR4 series, Joint Electron Device Engineering Council (JEDEC), the memory circuits standards defining body, added a new test called bit-error rate (BER) test. To cater to all these evolving test standards and requirements, high-speed testing equipment have dramatically improved in capacities over years.

Upgrades in T&M devices

T&M equipment have seen a number of transformations to scale up to the capabilities of DDR4 and higher versions and meet JEDEC standards, be it in terms of speed, size or channel support. Here are some of the major trends.

Higher speed, more parallel DUTs. DDR memory test setups are improving in speed

Before buying any equipment for high-speed memory testing, it is better to go through the tips and trends mentioned here